

AF0QT Practice Test (v2.1) – Questions

AF0QT-Style Comprehensive Practice Test (v2.1) – QUESTIONS ONLY

Instructions: Answer on a separate sheet. Timing and difficulty approximate the AF0QT format.

===== Section 1: Verbal Analogies (10) =====

1. COMPASS : DIRECTION :: CLOCK : ? A. time B. alarm C. battery D. minute hand
2. PRUDENT : CARELESS :: DILIGENT : ? A. lazy B. thorough C. eager D. early
3. OPAQUE : TRANSPARENT :: NOISY : ? A. silent B. loud C. echo D. music
4. GERMINATE : SEED :: IGNITE : ? A. fuel B. match C. heat D. smoke
5. ASTRONOMER : STARS :: GEOLOGIST : ? A. rocks B. maps C. fossils D. oceans
6. BENIGN : HARMLESS :: VIRULENT : ? A. mild B. contagious C. deadly D. common
7. PROTOTYPE : MODEL :: BLUEPRINT : ? A. plan B. building C. paper D. sketch
8. ECONOMIZE : SAVE :: SQUANDER : ? A. waste B. budget C. earn D. borrow
9. ALTITUDE : HEIGHT :: LATITUDE : ? A. north-south position B. temperature C. time D. speed
10. RUDDER : YAW :: AILERON : ? A. pitch B. roll C. thrust D. drag

===== Section 2: Word Knowledge (10) – Synonyms =====

1. TENABLE A. defensible B. fragile C. unlikely D. hostile
2. MALLEABLE A. rigid B. pliable C. brittle D. constant
3. INEPT A. capable B. skillful C. clumsy D. deft
4. LUCID A. dim B. clear C. frantic D. noisy
5. ABRIDGE A. lengthen B. compile C. shorten D. expand
6. COPIOUS A. scarce B. abundant C. tiny D. average
7. RELINQUISH A. seize B. ignore C. surrender D. hoard
8. MYRIAD A. few B. countless C. rare D. single
9. STOIC A. dramatic B. euphoric C. unemotional D. hostile
10. ELOQUENT A. shy B. inarticulate C. articulate D. speechless

===== Section 3: Reading Comprehension (10) =====

Passage A:

Effective crew communication reduces errors more than raw technical skill alone. Crew Resource Management (CRM)

trains teams to share workload, speak up about concerns, and cross-check critical steps. The goal is not hierarchy removal, but to create a climate where information flows to the person who needs it most, especially during high workload phases.

1. Main purpose of CRM? A. Replace pilots with automation B. Improve teamwork to reduce errors
 C. Reduce training hours D. Eliminate hierarchy
2. CRM encourages: A. Withholding concerns B. Silence in emergencies C. Cross-checks & speaking up D. Solo decisions
3. Tone toward CRM is: A. critical B. supportive C. indifferent D. humorous
4. Information should: A. Flow only upward B. Be documented, not discussed C. Go to whoever needs it D. Be restricted to maintenance
5. CRM is most vital during: A. Low workload cruise B. High workload phases C. Post-flight only D. Preflight paperwork

Passage B:

Cold fronts mark boundaries where colder, denser air undercuts warmer air, forcing it to rise. This lifting often produces lines of showers or thunderstorms and a wind shift, commonly from southerly ahead of the front to westerly or northwesterly behind it. After passage, temperatures usually drop and visibility can improve as drier air moves in.

6. Primary effect of a cold front: A. Warm air settles under cold B. Cold undercuts warm C. Winds calm to zero D. Humidity always increases
7. Weather along front often includes: A. Clear skies B. Lines of showers/storms C. Blizzards always D. Heat waves
8. Typical wind shift after passage: A. Southerly → W/NW B. Easterly → S C. N → S D. No change
9. Temperature after passage usually: A. rises B. falls C. unchanged D. spikes then stabilizes
10. Visibility after passage tends to: A. worsen B. improve C. remain poor D. become smoky

===== Section 4: Arithmetic Reasoning (10) =====

Generated 2025-10-23 – AF0QT Practice (v2.1)

1. Pump A fills a tank in 4 h, Pump B in 6 h. Together, time? A. 2.0 B. 2.4 C. 2.8 D. 3.0
2. \$250 discounted 20%, then taxed 8% on sale price. Final price? A. 216 B. 210 C. 220 D. 200
3. If x is 30% of y and y=150, x=? A. 45 B. 50 C. 40 D. 35
4. Car averages 54 mph for 3.5 h. Distance? A. 189 B. 175 C. 162 D. 200
5. Simple interest on \$500 at 4% for 3 years? A. 50 B. 60 C. 45 D. 40
6. Recipe: 3 c flour makes 12 cookies. For 18 cookies? A. 4 B. 4.5 C. 5 D. 3.5
7. Train covers 90 miles in 1.5 h. Avg speed? A. 50 B. 55 C. 60 D. 65
8. Price rises from 120 to 150. Percent increase? A. 20% B. 25% C. 30% D. 33%
9. Tank is $\frac{3}{5}$ full; add 24 gal to fill. Capacity? A. 50 B. 60 C. 70 D. 80
10. Buy 2 pens at 1.50 and 3 notebooks at 2.80. Total? A. 11.40 B. 12.20 C. 10.80 D. 13.00

==== Section 5: Math Knowledge (10) =====

1. Solve $5x+2=17$. A. 1 B. 2 C. 3 D. 4
2. Area of triangle with base 10 and height 7. A. 34 B. 35 C. 36 D. 37
3. Simplify $(3x^2y)(-2xy^3)$. A. $-6x^3y^4$ B. $-5x^2y^2$ C. $6x^3y^2$ D. $-6x^2y^3$
4. Slope of $y=-4x+5$. A. -5 B. 4 C. -4 D. 5
5. Factor $2x^2-8x$. A. $2x(x-4)$ B. $(x-4)(x-2)$ C. $2(x-4)$ D. $x(2x+8)$
6. 40% of 90 A. 36 B. 32 C. 38 D. 34
7. Right triangle legs 9 and 12. Hypotenuse? A. 13 B. 14 C. 15 D. 16
8. Median of [1,4,6,7,10,12,13] A. 6 B. 7 C. 8 D. 9
9. If $f(x)=2x^2-3x+1$, find $f(3)$. A. 8 B. 9 C. 10 D. 11
10. Volume of a cylinder, $r=3$, $h=5$, $\pi \approx 3.14$. A. 144.5 B. 141.3 C. 138.2 D. 150.0

==== Section 6: Table Reading (10) =====

City	Jan	Feb	Mar
Alpha	52	58	61
Bravo	60	57	65
Charlie	47	49	62
Delta	63	66	68
Echo	55	60	70
Foxtrot	59	56	64

1. Feb value for Delta? A. 64 B. 66 C. 68 D. 70
2. Highest Jan value belongs to: A. Alpha B. Bravo C. Delta D. Echo
3. Diff: (Mar of Echo) – (Jan of Charlie) A. 21 B. 23 C. 25 D. 27
4. Sum of Jan values for Alpha and Foxtrot A. 109 B. 110 C. 111 D. 112
5. Average of all Feb values (nearest integer) A. 56 B. 58 C. 60 D. 62
6. Which month is highest for Bravo? A. Jan B. Feb C. Mar D. Tie
7. Is Mar greater than Feb for Echo? A. Yes B. No
8. Lowest Mar value is in: A. Alpha B. Bravo C. Charlie D. Foxtrot
9. (Jan of Delta) + (Mar of Charlie) = ? A. 123 B. 125 C. 127 D. 129
10. For Foxtrot, Jan exceeds Feb by: A. 1 B. 2 C. 3 D. 4

===== Section 7: Instrument/Heading (10) =====

1. 045°; turn right 120°. New heading? A. 135 B. 165 C. 225 D. 315
2. 270°; turn left 90°. New heading? A. 000 B. 090 C. 180 D. 360
3. 315°; turn right 60°. New heading? A. 015 B. 045 C. 075 D. 105
4. 120°; turn left 150°. New heading? A. 330 B. 300 C. 270 D. 090
5. 200°; turn right 170°. New heading? A. 010 B. 030 C. 350 D. 020
6. 010°; turn left 45°. New heading? A. 325 B. 335 C. 355 D. 045
7. 085°; turn right 275°. New heading? A. 000 B. 085 C. 180 D. 360
8. 350°; turn left 200°. New heading? A. 140 B. 150 C. 160 D. 200
9. 090°; turn right 225°. New heading? A. 225 B. 270 C. 315 D. 045
10. 180°; turn left 270°. New heading? A. 090 B. 270 C. 000 D. 045

===== Section 8: Block Counting (10) – Sum the stack heights =====

1. 4-4-4-4 A. 14 B. 15 C. 16 D. 18
2. 1-3-5-3-1 A. 11 B. 12 C. 13 D. 15
3. 2-5 A. 5 B. 6 C. 7 D. 8
4. 6-2-6 A. 12 B. 13 C. 14 D. 15
5. 7 A. 5 B. 6 C. 7 D. 8
6. 3-3-1-2-4 A. 12 B. 13 C. 14 D. 15
7. 5-5-1 A. 9 B. 10 C. 11 D. 12
8. 2-2-2-2-2-2 A. 12 B. 13 C. 14 D. 16
9. 1-2-3-4-5 A. 14 B. 15 C. 16 D. 17

10. 9–1 A. 9 B. 10 C. 11 D. 12

===== Section 9: Aviation Information (10) =====

1. Angle of attack is the angle between the: A. wing & horizon B. chord line & relative wind
C. fuselage & airflow D. chord line & ground
2. To hold altitude in a steeper bank, a pilot generally must: A. reduce AoA B. increase AoA/add power
C. reduce power only D. push forward on yoke
3. Detonation is: A. abnormal, near-explosive combustion in the cylinder B. carburetor icing
C. magneto failure D. same as preignition
4. High density altitude typically makes takeoff distance: A. shorter B. longer C. unchanged
D. variable but usually shorter
5. Elevator trim is used primarily to: A. turn the aircraft B. relieve control pressure C. increase lift
D. deploy flaps
6. Best-angle climb speed V_x is, relative to V_y : A. less than V_y B. greater than V_y C. equal to V_y
D. unrelated
7. Which instrument uses both pitot and static pressure? A. Altimeter B. VSI C. Airspeed indicator
D. Magnetic compass
8. A forward CG generally: A. increases stability & increases stall speed B. decreases stability & decreases stall speed
C. increases stability & decreases stall speed D. no effect
9. Wake turbulence is strongest when the generating aircraft is: A. heavy, clean, slow B. light, dirty, fast
C. heavy, dirty, fast D. light, clean, slow
10. A standard-rate turn equals: A. $1^\circ/\text{s}$ B. $2^\circ/\text{s}$ C. $3^\circ/\text{s}$ D. $4^\circ/\text{s}$

===== Section 10: Situational Judgment (10) – BEST response =====

1. Preflight reveals a discrepancy outside MEL limits. A. Depart anyway B. Stop; coordinate maintenance per MEL and delay if needed
C. Ignore; note later D. Ask ramp crew to sign it off
2. Teammate's critical task is slipping; causes unclear. A. Assign blame B. Meet to surface blockers; rebalance & document plan
C. Remove them from task D. Escalate immediately without context
3. You made an error in a brief. A. Hide it B. Own it; correct and share lessons C. Blame data
D. Minimize it
4. Two leads give conflicting priorities. A. Pick one arbitrarily B. Seek joint clarification & confirm priorities
C. Delay indefinitely D. Split the difference silently

5. Safety issue may delay schedule. A. Proceed anyway B. Raise concern; propose mitigations & options C. Bury the finding D. Shift blame
6. Quiet new member seems disengaged. A. Exclude them B. Check in 1:1; pair with mentor; set small wins C. Critique publicly D. Remove tasks
7. Stakeholder adds last-minute scope. A. Agree to all B. Clarify requirements; negotiate tradeoffs & impacts C. Refuse rudely D. Ignore request
8. Morale dips after setbacks. A. Punish publicly B. Hold blameless retro; celebrate small wins; reset goals C. Cancel meetings D. Ignore it
9. You suspect data quality issues in a nearly finished analysis. A. Ship it anyway B. Flag risk; triage checks; document assumptions C. Delete the work D. Blame the source
10. A teammate is consistently late to stand-ups. A. Publicly call out B. Private check-in on causes & expectations; agree on fixes C. Ignore D. Report immediately

==== Section 11: Physical Science (10) =====

1. Which quantity is a vector? A. speed B. mass C. velocity D. temperature
2. Newton's third law states that action-reaction forces act on: A. the same object B. different objects C. the ground only D. massless objects
3. If wave speed stays constant while frequency increases, wavelength: A. increases B. decreases C. stays the same D. becomes undefined
4. Electrical power in a DC circuit is given by: A. $V+I$ B. V/I C. VI D. $V-I$
5. Which phase change requires energy input? A. freezing B. condensation C. evaporation D. deposition
6. A solution with pH 3 compared to pH 5 is: A. 2× more acidic B. 10× more acidic C. 20× more acidic D. 100× more acidic
7. The particle that determines an element's identity is the: A. neutron B. proton C. electron D. positron
8. A 10 Ω resistor carries 2 A. The voltage across it is: A. 5 V B. 10 V C. 20 V D. 0.2 V
9. The SI unit of energy is the: A. newton B. joule C. watt D. pascal
10. A car travels at constant speed in a circle. Its acceleration points: A. zero (none) B. outward C. toward center D. tangent to path